

Complex-Valued Models Improve EEG Behavior Decoding

A Thesis Presented

by

Guillermo A. Mendoza Franco

To the Keck Science Department

of

Claremont McKenna, Scripps, and Pitzer Colleges

In Partial Fulfillment of

The Degree of Bachelor of Arts

Senior Thesis in Computational Neuroscience

April 27th, 2026

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Abstract

Oscillatory signals, like those captured in electroencephalograms (EEGs), can carry information through patterns of phase and amplitude across electrodes. These patterns can be identified using a Hilbert transformation of the signal, resulting in complex-valued signal representations. Our lab has developed a neural network, known as TIMBRE (Tracking Informative Multivariate Brain Rhythms Efficiently), capable of learning these behaviorally informative patterns from multi-channel brain wave recordings (Agarwal et al., 2026). Unlike traditional models, TIMBRE learns complex-valued projections of multivariate signals and implicitly encodes phase relationships, allowing it to outperform benchmark models like Common Spatial Patterns (CSP) when applied to hippocampal local field potentials (LFPs) known as theta rhythms. While CSP is typically applied to real-valued signals (rCSP), we have adapted the algorithm to function on complex-valued signals (cCSP).

At its core this study benchmarks complex-valued models with real valued ones on a motor imagery decoding and classification task. We found that methods leveraging complex-valued signals outperformed rCSP across the board on data from a motor imagery task. When applied to this EEG dataset (BCI Competition III dataset IVa, 100 Hz sampling rate), TIMBRE remained competitive but did not decisively outperform cCSP overall, departing from expectations informed by performance on rodent hippocampal LFPs (Agarwal et al. 2026). Extensive hyperparameter tuning was performed on TIMBRE and cCSP, which led to moderate accuracy gains across subjects for TIMBRE, but minimal changes for cCSP. Spectral analysis of TIMBRE's pre-magnitude activations revealed a class-dependent effect on log power that was statistically significant in mu (8–13 Hz) and beta (13–30 Hz) frequency bins after Benjamini–Hochberg FDR correction ($p < 0.05$; class effect from a two-way ANOVA, power ~

class + subject, with subject as a blocking factor). 'Foot'-labeled trials drove higher activation power across these bins. Thus, beyond classification performance, models such as TIMBRE can be used to arrive at sometimes unexpected physiological understanding.

Introduction

Electroencephalography (EEG) allows researchers to measure brain activity at large through a signal that encodes behaviorally relevant information in oscillatory structure distributed across electrodes and frequencies (Kaplan & Zimmer, 2020; Sihn & Kim, 2025; Westlin et al., 2023). One well-known example is motor imagery, where imagined movements are reflected in changes to sensorimotor rhythms in the mu ($\sim 9\text{--}12$ Hz) and beta ($\sim 13\text{--}30$ Hz) bands. Decoding oscillatory neural activity remains a central problem in computational neuroscience as it underpins applications in brain–computer interfaces (BCIs), cognitive-state inference, and neural dynamics with the potential to improve clinical decision making and post-treatment quality of life.

A well-established and longstanding approach to this problem is Common Spatial Patterns (CSP) which learns spatial filters that maximize variance differences between classes by solving a generalized eigenvalue problem over class-specific covariance matrices (Koles et al., 1990; Wang et al., 2004). The classical CSP formulation, including the implementation used here, is binary; in this dataset the two classes are right-hand and right-foot motor imagery. Because the generalized eigenvalue formulation directly optimizes the spatial filters for class discriminability, CSP stands as a mathematically principled method with outputs tied to the underlying signal structure rather than a mere benchmark. It is computationally lean, interpretable, and remains a widely-accepted baseline for motor imagery classification, as literature shows that it directly captures event-related desynchronization in sensorimotor rhythms (Pfurtscheller & Neuper, 1994; Mishuhina & Jiang, 2021). A large portion of EEG-based BCIs continue to rely on CSP and its variants because spatial filters can be fit per subject from a short calibration block, after which classification is fast enough for online use (Ang et al., 2012). The expensive step is the

per-subject training; deployment is cheap. Furthermore, CSP often matches (and even exceeds) the performance of more complex models, cementing it as the golden standard for validating new methods.

But if CSP is *that* good, why would anyone bother looking at different classification methods?

Canonically, CSP operates on real-valued signals which simplifies oscillatory activity to band-limited power, thereby discarding phase information and constraining the representation of neural dynamics. EEG oscillations are more naturally described as analytic signals where instantaneous amplitude, also known as the envelope, and phase represent the process at hand. Real-valued pipelines do not make phase inaccessible, as it can be carried as a pair of real channels (amplitude and unwrapped phase). The issue is that the standard real-valued CSP objective operates on band-limited power and so does not directly use phase or the cross-electrode complex covariance structure that the analytic signal exposes. Empirical work that does feed phase or full complex covariance into the classifier reports accuracy gains over amplitude-only features (Akuthota et al., 2023; Malan & Sharma, 2022; Mishuhina & Jiang, 2021). Mishuhina & Jiang in particular implemented this by computing CSP on the time–frequency-decomposed analytic signal, which is the closest existing analogue to the cCSP we test here. This motivates moving the spatial-filter problem itself into the complex domain. A key advantage of this representation domain is that it makes the full oscillatory structure of the signal directly accessible without requiring explicit phase unwrapping or time-frequency decomposition. Obtained via the Hilbert transform, analytic signals encode amplitude and phase in a unified regime. When implemented with this domain in mind, CSP – or complex CSP (cCSP) – is able to create spatial filters that capture phase relationships across electrodes on top of amplitude structure. In general, complex-valued models tap into the wealth of features EEG

has to offer due to its oscillatory nature, this providing a mechanism to represent traveling waves, phase synchronization (or lack thereof), and other spatiotemporal phenomena that are difficult to express through real-valued pipelines (Grabot et al., 2025; Malan & Sharma, 2022). Despite these advantages, clarity on extending complex-valued methods to scalp EEG is scarce. Prior work exhibits gains from complex-valued CSP pipelines (Mishuhina & Jiang, 2021), but does not establish a clear framework for applying these methods to standard motor imagery data, or understanding the features obtained. At the same time, many modern approaches have let classification accuracy take precedence over interpretability, leaving open the question of what neural dynamics drive decoding performance. Broader surveys of EEG-based BCIs support the existence of this gap, as performance improvements through architectural modifications or abstracted feature engineering fail to link model outputs to neurophysiology (Akuthota et al., 2023; Ang et al., 2012).

This gap is of particular interest for biomedical applications, where **both** performance and interpretability are required. Accuracy by itself is not enough if model outputs cannot be related back to the very processes that are placed under the microscope, namely oscillatory rhythms, spatial organization, and frequency-specific activity. An engineering-first approach necessitates models that perform well and expose the structure of the underlying signal.

In this study we evaluated complex-valued methods for EEG motor imagery classification, with a particular focus on complex CSP (cCSP) and TIMBRE (Agarwal et al., 2026). TIMBRE is a complex-valued neural network whose architecture is designed around the structure of oscillatory data: it operates directly on analytical signals and extracts features as complex-valued projections and extracts magnitude features. cCSP and TIMBRE therefore share an input domain and an output structure (complex projection $\rightarrow$ magnitude $\rightarrow$ linear classifier), and differ only in

how the projection is obtained analytically, by solving a generalized eigenvalue problem, or by gradient descent. This pairing is the empirical question of this thesis: does adding learned flexibility on top of the complex-valued formulation buy anything once the analytic signal is already in play, or does the closed-form solution already capture the discriminative structure? cCSP plays the role of the principled analytic baseline within the complex domain; rCSP anchors the comparison to the long-standing real-valued benchmark. Given its history in the field, CSP was used as a baseline to anchor comparisons (*BCI Competition III: Results*, 2003).

Hyperparameter optimization was used to explore the limitations of each model class, assessing whether added complexity yields meaningful gains over well-optimized baselines. This work aims to show that complex-valued representations improve classification performance, and interpretability of learned features. By analysing model activations in the frequency domain, we link decoding performance to specific oscillatory rhythms and demonstrate how complex-valued models capture information that their real-valued counterparts cannot, positioning them as a principled extension of classical spatial filtering for EEG.

Methods

Dataset

EEG data were obtained from BCI Competition III Dataset IVa (*Data Set IVa for the BCI Competition III*, 2003). Recordings consist of 118-electrode scalp EEG acquired during a motor imagery task in which subjects imagined movements of the right hand or foot. The original sampling rate of 1000 Hz was downsampled to 100 Hz for all analyses. Trials were structured with event markers indicating onset and were accompanied by a label matrix encoding trial index, class label (class 1 or 2, or `NAN`), and a binary validity flag; validity was set to 1 for trials with class labels. Only valid trials were retained. Each trial spans 350 samples at 100 Hz (3.5 seconds).

Three of the five subjects (aa, al, aw) were retained for analysis. Selection was based on the number of valid trials available, not random sampling, so quantitative results are reported per subject and the work makes no claim about generalization across the full subject pool.

Preprocessing

All preprocessing steps follow the notebook implementation (Appendix).

Filtering & Analytic Signal Construction

Raw EEG signals were high-pass filtered at 2 Hz using a zero-phase Butterworth filter to remove slow drifts. The Hilbert transform was applied to each filtered signal to construct complex-valued analytic signals:

$$x_a(t) = x(t) + i \cdot \mathcal{H}[x(t)]$$

where $\mathcal{H}[x(t)]$ denotes the Hilbert transform. Equivalently, the analytic signal is constructed by zeroing the negative-frequency components of $x(t)$ in the Fourier domain and inverse-transforming back, which yields a signal whose modulus is the instantaneous envelope

and whose argument is the instantaneous phase. Both operations were performed in a single step. The resulting analytic signals encode the envelope and phase as the modulus and argument of the complex representation, respectively.

Trial Selection

Filtered samples were masked to retain valid trials using the aforementioned validity flag. The raw label column encodes class as 1 (right hand) or 2 (right foot); subtracting 1 yields the zero-indexed labels used for downstream calls in the code. The resulting class assignments were: “Class 0 = hand”, “Class 1 = foot”.

Whitening

Whitening decorrelates electrodes and rescales each component to unit variance, replacing the input covariance with the identity. This is the standard preconditioning step for CSP without it, the generalized eigenvalue problem reduces to comparing class covariances against a poorly-conditioned overall covariance, and channel-level amplitude differences dominate the filters. The same whitened input is used by TIMBRE so that all three methods see the same preprocessed signal and any performance differences are attributable to the projection step. Whitening was fit per cross-validation fold using training indices only and then applied to the full data, so test-fold samples never inform whitening parameters. The whitened output is complex-valued (it was performed on the analytic signal).

Feature Extraction

Real-Valued CSP (rCSP)

The real part of the whitened analytic signal was used as input for rCSP. Class specific covariance matrices were estimated from the training data and spatial filters were obtained by solving the generalized eigenvalue problem per class against pooled covariance:

$$\Sigma_i w = \lambda \Sigma w$$

where Σ_i is the per-class covariance. The top eigenvectors per class were selected as spatial filters. Features were computed as the magnitude of the projected signals:

$$f = |W^\top X|$$

These magnitude features were passed directly to the logistic regression classifier.

Complex-valued CSP (cCSP)

cCSP used the full complex-valued whitened signal as input. It followed the same generalized eigenvalue formulation in the complex domain. Spatial filters were obtained per class against the complex pooled covariance; the conjugate of the selected filters was applied to the data during projection. Features were the elementwise magnitude of the complex projected signals, which preserved amplitude information from the analytic representation, and discarded explicit phase.

These features were passed to logistic regression.

CSP spatial filters were computed for $n \in \{1, 2, 4, 8\}$ components per class for both real-valued and complex CSP.

Models

Logistic regression

A logistic regression classifier with L2 penalty (`LogisticRegression(penalty=L2, max_iter=1000)`, scikit-learn) served as the downstream classifier for both rCSP and cCSP feature vectors. A separate model instance was fitted per fold.

TIMBRE

TIMBRE (Agarwal et al., 2026) is a shallow complex-valued neural network originally developed for and validated on carrier-free decoding of multi-electrode LFP oscillations. The architecture learns complex-valued spatial projections of the input signal and extracts

behaviorally informative features via magnitude extraction, without the need for a reference oscillation. This work applies TIMBRE to scalp EEG motor imagery classification, using the same principles in a new domain.

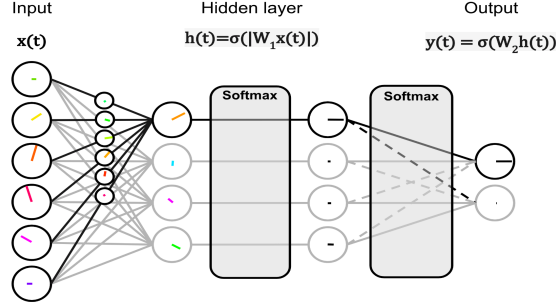


Figure 1. TIMBRE architecture, consisting of a hidden layer that takes a complex-valued linear projection of the inputs.

TIMBRE takes the concatenated real and imaginary parts of the whitened signal as input.

The network consists of a complex-valued linear projection layer with H hidden units followed by magnitude extraction and a softmax activation over the hidden layer. When $H > 0$, the softmax-activated hidden activations are passed to a fully connected output layer that yields class probabilities. When $H = 0$ (no hidden layer), the complex projection maps directly to the number of classes with no intermediate hidden layers. The network is trained with Adam and categorical cross-entropy loss.

TIMBRE was evaluated with hidden layer sizes $H \in \{0, 4, 8, 16\}$.

Training & Evaluation

Cross-validation

All models were evaluated under 5-fold crossvalidation applied per subject. Folds were assigned trial by trial, enforcing balance class representation across folds. Test indices for each fold were

held out in full and used only for final evaluation. Whitening was computed from training indices only and applied to the full dataset; test data had no influence on preprocessing parameters.

TIMBRE Training

Initially, TIMBRE was trained using the Adam optimizer with early stopping on validation loss using predetermined hyperparameters (learning rate= 1×10^{-3} , patience=1). Maximum training epochs were set to 100. Reported accuracy is the validation accuracy at the final training epoch. After the introduction of Optuna search, all tuned models used `max_epochs = 120` to give the optimizer headroom; this difference is noted because the untuned baseline therefore caps slightly earlier than tuned runs.

Hyperparameter Optimization

Hyperparameters for TIMBRE were explored via two methods. Initially, a comprehensive grid search was conducted over discrete and continuous parameter spaces (number of hidden nodes, learning rate, batch size, regularization settings), with all configurations trained to completion without early stopping. This approach was time consuming, and though it yielded results they failed to meet standards (optimization was ineffective due to computational constraints).

Thus, an alternative approach was adopted. Optuna (Akiba et al., 2019), an automated optimization framework, was used on TIMBRE and cCSP. Each optimization used a TPE sampler, ran for 100 trials per fold, per subject, and nested inner validation.

For TIMBRE, the search explored hidden-nodes, learning rate, magnitude scale, regularization, and normalization strategy. Each Optuna study (one study per outer fold per subject per model) was persisted to a SQLite backend to enable post-hoc analysis of the hyperparameter landscape. For each outerfold, an inner validation split was derived from the outer training data. The

objective function maximized validation accuracy. Optuna's TFKerasPruningCallback was attached to each Keras training run, monitoring validation accuracy. After each epoch, Optuna's pruner compares the trial's intermediate value to the empirical distribution of values from completed trials at the same epoch; trials in the lower tail are stopped early to redirect compute toward more promising configurations. The sampled hyperparameter space for TIMBRE is described in Table 1. Continuous, scale-spanning parameters (learning rate, L2 penalty, logistic-regression C, logistic-regression tolerance, ridge regularization) were sampled log-uniformly. Discrete parameters with a small number of meaningful values (hidden nodes, batch size, magnitude scale, normalization strategy, regularization strategy, solver, number of spatial components) were sampled categorically. A consequence of categorical sampling is that the per-parameter sensitivity plots for those variables (Figure 3, right three panels) display column-of-points patterns rather than the smooth point clouds visible for the log-uniform parameters (Figure 3, left two panels); a continuous parameterisation of magnitude scale and batch size could be considered in future work.

Table 1. Hyperparameter space sampled for TIMBRE using Optuna.

Hyperparameter	Range/Values	Sampling
Hidden Nodes	{0, 4, 8, 16}	Categorical
Learning rate	$[1 \times 10^{-5}, 3 \times 10^{-2}]$	Log-uniform
Batch size	{16, 32, 64, 96, 128}	Categorical
Magnitude scale	{0.25, 0.5, 1.0, 2.0, 4.0}	Categorical
Normalization strategy	{unit-norm, batch normalization}	Categorical
Regularization strategy on weights	{L2, no regularization}	Categorical
L2 penalty	$[1 \times 10^{-6}, 1 \times 10^{-1}]$	Log-uniform
Patience	[1, 2, 3, 5, 8]	Categorical

For CSP the search included ridge regularization strength, number of spatial components, and logistic regression regularization. Like with TIMBRE, Optuna studies were written to a SQLite backend to support post-hoc analysis of the hyperparameter landscape. For each outerfold, an inner validation split was derived from the outer training data and the objective function maximized validation accuracy. The sampled hyperparameter space for CSP is summarized in Table 2.

Table 2. Hyperparameter space sampled for CSP using Optuna.

Hyperparameter	Range/Values	Sampling
Number of spatial components	$\{1, 2, 4, 8\}$	Categorical
Ridge regularization	$[1 \times 10^{-6}, 1 \times 10^{-1}]$	Log-uniform
Logistic regression regularization (C)	$[1 \times 10^{-4}, 1 \times 10^3]$	Log-uniform
Logistic regression solver	$\{\text{lbfgs}, \text{liblinear}\}$	Categorical
Logistic regression tolerance	$[1 \times 10^{-6}, 1 \times 10^{21}]$	Log-uniform

The best hyperparameters from each inner search were used to retrain the model on the full outer training fold, evaluating performance on the outer held-out test set. Hyperparameter optimization was used to identify potential performance ceilings, not to maximize standalone benchmark performance.

For interpretability analyses, a single hyperparameter configuration was obtained by aggregating the fold-wise optimal configurations (selected via inner validation), and models trained with this configuration were used only for analysis and not for performance evaluation.

Evaluation Metric

Classification performance was measured as sample-level accuracy: the proportion of held-out time-samples whose predicted class matched the trial label. Predictions are made independently per time-sample (not per trial); each 350-sample trial therefore contributes 350 predictions, all

sharing the trial's class label. Reported accuracies are the mean across the five outer folds, per subject, with the standard deviation across folds shown as error bars in the figures.

Spectral Analysis

To understand what oscillation each model responds to, power spectra were computed from the internal activations of both TIMBRE and cCSP on held-out test data. A power spectrum decomposes a time series into the energy contributed by each frequency component; concentration in a frequency band indicates that the projection is selectively responsive to oscillations in that band.

For TIMBRE, activations were extracted from the complex-valued projection layer, before magnitude extraction across all five test folds, then pooled. For cCSP, the complex projected signals prior to the magnitude operation were used. Examining pre-magnitude signal allows us to capture the full frequency spectrum of what the model has learned to project onto.

Per class power spectra were computed by segmenting each activation series into non-overlapping windows of 350 samples (3.5 s at 100 Hz, one trial), computing the squared magnitude of the Fast Fourier Transform per segment, and averaging across segments. This was performed separately for each class label. Per-class power spectra were computed by segmenting each pre-magnitude activation series into non-overlapping 350-sample windows (3.5 s at 100 Hz, one trial), taking the squared magnitude of the per-segment FFT, and averaging across segments within each class. For visualization, results are reported as the log-power difference between the foot and hand classes:

$$\Delta \log\text{-power}(f) = \log(1 + P_{\text{class } 1}(f)) - \log(1 + P_{\text{class } 0}(f))$$

so that values above zero indicate a frequency where foot-imagery trials drive higher activation power than hand-imagery trials, and values below zero indicate the reverse.

Statistical significance of the class effect was assessed at each frequency bin with a two-way ANOVA on the per-trial log-power, with class as the factor of interest and subject as a blocking factor:

$$\log(1 + P_{\text{trial},f}) = \beta_0 + \beta_{\text{class}} \cdot \text{class} + \beta_{\text{subj}} \cdot \text{subject} + \varepsilon$$

The reported p-value at each frequency bin is the class-effect p-value. P-values were corrected for multiple comparisons across frequency bins using the Benjamini–Hochberg false-discovery-rate procedure. Subject is included only as a blocking term; its main effect is not reported because the model treats inter-subject differences as nuisance variance, and the design is the same set of trials across all units, so the subject p-value is not what the analysis is testing.

Results

Classification Performance

Performance curves across feature counts were generated using a fixed hyperparameter configuration obtained by aggregating top-performing Optuna configurations. These curves answer the question: holding the configuration fixed, how does accuracy scale as we vary the number of components? They are summary comparisons across model classes, not independent nested-CV performance estimates; the latter come from the per-fold Optuna runs reported in Hyperparameter Optimization. The motivation for separating these is that without a fixed configuration, accuracy differences across feature counts are confounded by the optimizer making different per-fold choices at each setting.

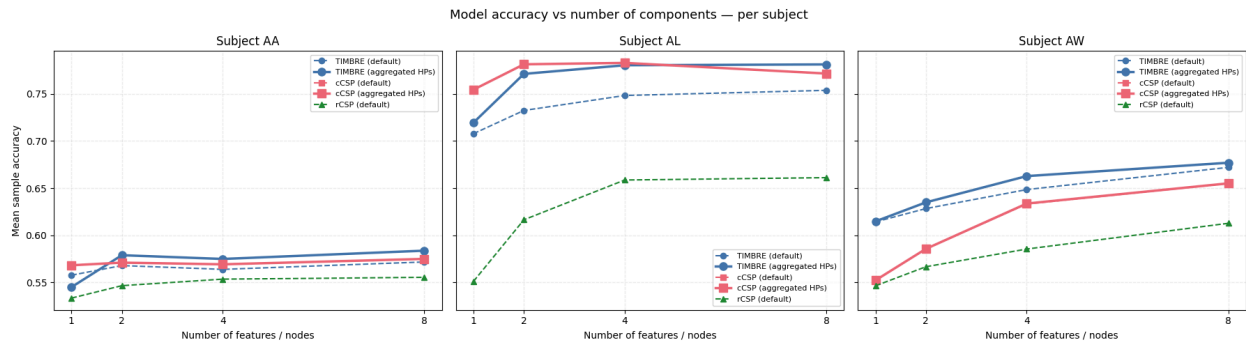


Figure 2. Mean model accuracy as a function of number of components per subject with standard deviation bars calculated across folds. Aggregated curves use post-hoc aggregated hyperparameters from prior tuning analyses. Note the perfect overlap between cCSP and its tuned version.

Baseline (rCSP + Logistic Regression)

Real-valued CSP combined with logistic regression serves as the primary baseline. Chance accuracy on this binary task is 0.5. rCSP achieves mean accuracies in the 0.54–0.61 range across subjects and component counts, which is consistent with previously reported numbers for this

dataset (Mishuhina & Jiang, 2021; Wang et al., 2004) and reflects the well-documented difficulty of BCI Competition III IVa: subjects had limited training (only 280 calibration trials) and the test labels in the original competition were withheld for blind evaluation, making absolute accuracies systematically lower than on more recent, larger-calibration BCI datasets.

Complex CSP

cCSP consistently outperforms real-valued CSP across all subjects and feature counts.

Improvements are apparent in all subjects, at all values of n tested, with the performance gap between cCSP and rCSP persisting across folds, with performance scaling more than linearly in the number of components.

Untuned TIMBRE

Prior to hyperparameter optimization, TIMBRE was evaluated with default settings across number of hidden nodes (learning rate: 1×10^{-3} , patience: 1), and on average, it produced comparable results to cCSP across subject and number of components.

Hyperparameter Optimization

For TIMBRE, across all subjects and folds, the optimizer consistently selected the number of hidden nodes, batch normalization paired with L2 regularization as the preferred configuration, due to slight gains relative to unit-norm. Subject-level unique configuration counts explored were 259 (al), 275 (aw), and 299 (aa).

For CSP, the optimizer selected the number of spatial components, ridge regularization, logistic regression regularization constant, logistic regression solver, and logistic regression tolerance.

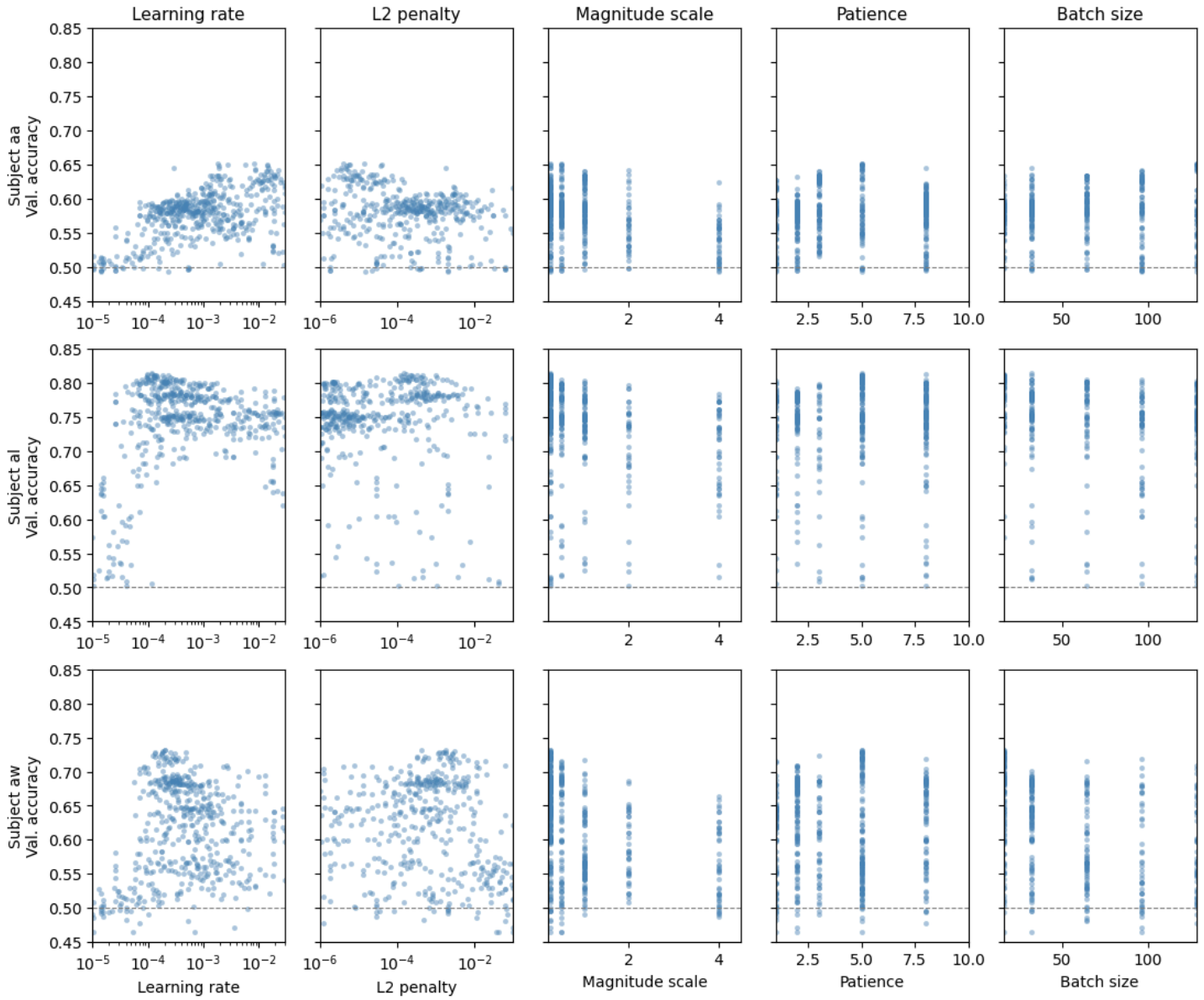


Figure 3. Optuna sensitivity plots showing inner-val accuracy as a function of learning rate, L2 penalty, magnitude scaling, patience and batch size for each subject. Point clouds illustrate subject-specific hyperparameter sensitivity and regions of stable or unstable TIMBRE performance across the search space.

The left two panels (learning rate, L2 penalty) display point clouds because both were sampled log-uniformly. The right three panels (magnitude scale, patience, batch size) display vertical

columns because these were sampled categorically; future hyperparameter searches in this pipeline could replace the categorical samplers with log- or linearly-uniform continuous samplers to reveal smoother sensitivity surfaces. The learning-rate and L2-penalty panels exhibit a near-mirrored shape across most subjects: configurations with high learning rate combined with high L2 penalty cluster at the bottom of the accuracy axis, while configurations with intermediate learning rate and low-to-moderate L2 cluster at the top. This is consistent with the two parameters interacting through the effective step size on the regularised loss, a high learning rate with strong L2 over-corrects each update, while an intermediate learning rate with mild L2 stabilises the trajectory.

Gains over untuned models

Gains were substantial for subject al and modest for subject aa, indicating that TIMBRE’s effective performance ceiling is subject-dependent, and optimization is effective in yielding results approaching best model subject level performance.. See figure 2.

cCSP exhibited minuscule gains (on the order of 1×10^{-3}) from optimization, indicating that analytically derived spatial filters already approximate an optimal solution to the objective.

Hyperparameter transferability

To assess whether optimized configurations encode subject-general structure or overfit to the subject on which they were tuned, top-performing configurations from each source subject were applied without retuning to all three target subjects. The best 10 optimized configurations from each subject were evaluated against the default parameter baseline on the same target subject and fold. Transfer deltas ($\Delta = \text{transfer accuracy} - \text{baseline accuracy}$) were small in magnitude, ranging from -0.015 to $+0.010$ across subject pairs. The interpretation depends on perspective. From the standpoint of *generalization*, this is a positive result: configurations optimized on one

subject neither help nor harm performance on another, so the per-subject Optuna procedure is not learning subject-idiosyncratic settings that would damage held-out subjects. From the standpoint of *gains over the default*, this is the limiting case: optimization gains do not transfer either, which means the within-subject improvements observed for al and aw (see *Gains over untuned models*) reflect tuning to that subject's data distribution rather than the discovery of a globally better configuration. Both readings are consistent with there being no single hyperparameter setting that is reliably better than the default across this three-subject pool.

The final hyperparameters for were obtained by aggregating the top 5 performance configurations from the hyperparameter search. Continuous parameters were estimated via the geometric mean, and discrete hyperparameters were selected via the mode. This aggregation reduces sensitivity to individual trial noise without abandoning the high-performing region of the search space.

Hyperparameter Sensitivity

Depicted in figures 2 and 3..

Learning rate was the parameter with the strongest visible relationship to inner validation accuracy across subjects: intermediate values ($\sim 10^{-4}$ to 10^{-3}) consistently produced higher scores, while very high rates degraded performance. L2 penalty showed a weaker and less consistent effect. Magnitude scale preferred lower values in the majority of top-performing configurations. Patience and batch size exhibited subject-specific high performance ranges.

Hidden node count affected performance to a large extent, but did not dominate performance within the optimization framework; the optimizer returned hidden nodes 0 and 16 in comparable proportions for subject aa, suggesting that for this subject the model capacity is not the binding constraint.

Spectral Analysis (Power plots)

Figures 3 and 4 show the activation spectra of TIMBRE and CSP, respectively. Power spectra of model activations revealed a consistent asymmetry in class-related responses across subjects.

Across both TIMBRE and cCSP, Class 1 (foot) was more strongly expressed in the learned projections, leading to a “dormant” unit. This held qualitatively across all three subjects.

The difference curve departs from zero in the mu and beta frequency bands, indicating that one class preferentially drives activations at the frequencies associated with sensorimotor rhythms.

Where the difference is predominantly positive across a broad frequency range, Class 1 accounts for the majority of activation power in that unit/component. The reverse pattern was not observed as often, and occurred with a weaker magnitude.

The lower panels on TIMBRE’s activation spectra (Figure 3) show the statistical significance of class dependent differences in activation power spectra, assessed using a two-way ANOVA (power $\sim$ class + subject) at each frequency bin. p-values for the class effect were corrected for multiple comparisons using the Benjamini–Hochberg FDR procedure. The corrected p-value for class effect is plotted as $-\log_{10}(p_f^{\text{FDR}})$. Red regions indicate instances where class differences are statistically significant ($p < 0.05$, FDR-corrected). This test found activation differences across all subjects. cCSP features are trial-level variance features, so frequency-resolved analysis is not applicable.

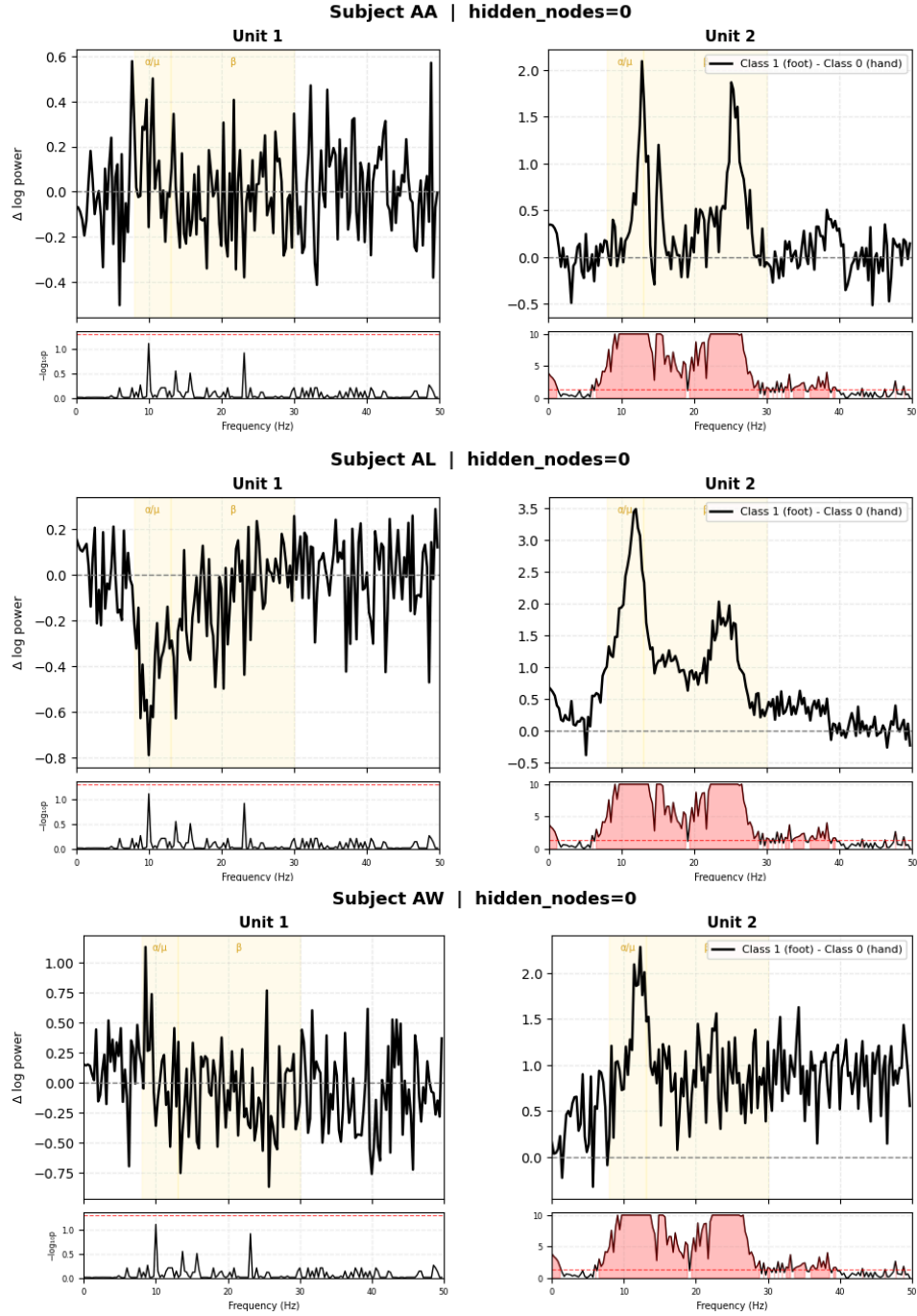


Figure 3. Per-subject difference power spectra for fixed-config TIMBRE ($H = 0$) with μ and β bands annotated. The lower strip in each panel is the $-\log_{10}$ of the BH-FDR-corrected class-effect p-value from a two-way ANOVA (power $\sim$ class + subject) fit once on the trial-level pooled data across all three subjects; subject is treated as a blocking term, and only the

class-effect p-value is shown. Because the same pooled ANOVA produces the p-vector, the significance strip is identical across the three subject panels by construction; the per-subject spectra above each strip differ. Each panel labelled 'Unit k' corresponds to the k-th hidden unit of TIMBRE. For TIMBRE with $H = 0$ (no hidden layer), the single 'Output projection' panel corresponds to the $H \times 2 = 2$ -channel output projection that maps directly to the class probabilities.

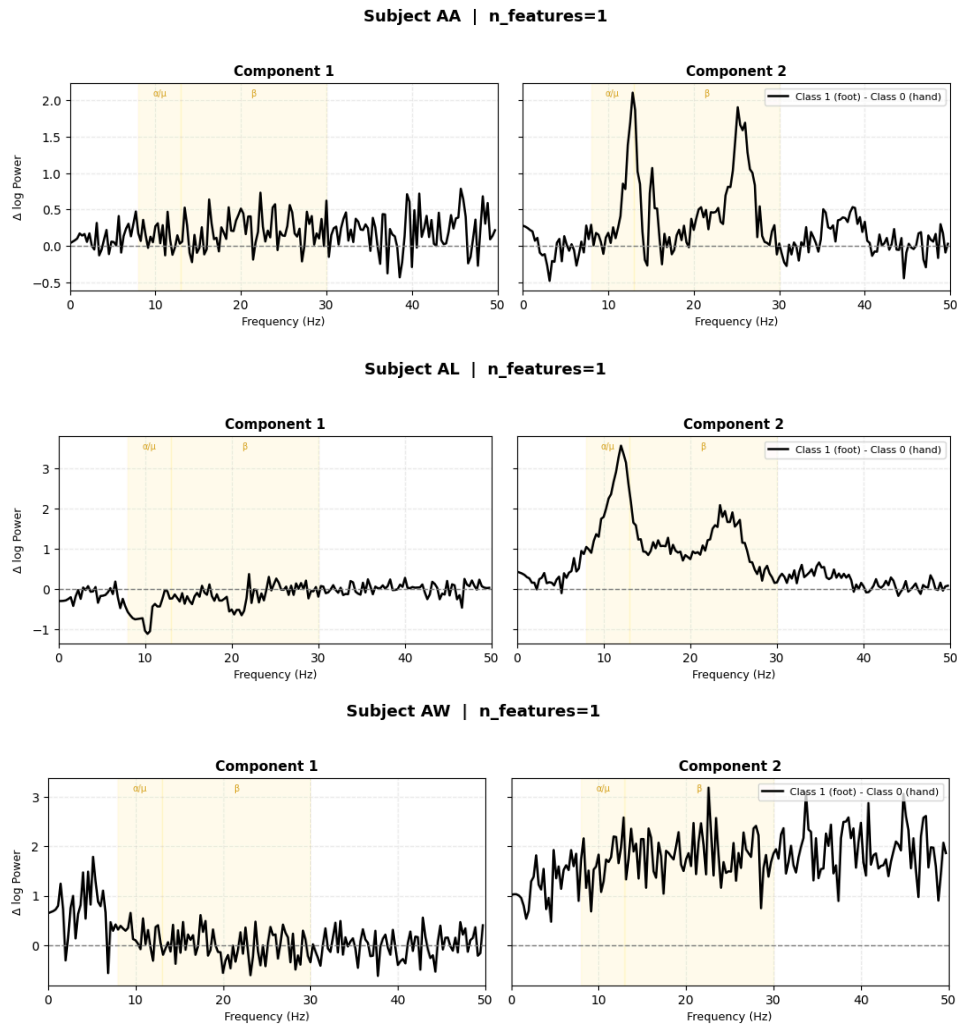


Figure 4. Per-subject difference power spectra for fixed-config cCSP (1 component) with bands μ and β annotated. Each panel labelled 'Component k' corresponds to the k-th retained spatial filter of cCSP.

The full per-subject power spectra for both TIMBRE and cCSP can be found in additional figures.

Discussion

Complex-valued representations improve EEG motor imagery decoding

The most consistent finding in this thesis is that operating in the complex domain is more advantageous than real-valued spatial filtering. cCSP outperformed rCSP across all three subjects and across the full range of component counts tested, with gains present at every value of n tested. This is not an artifact of optimization choices, as both pipelines used the same whitened analytic signal, generalized eigenvalue formation, and logistic regression classifier. The only difference between the two was whether the covariance structure was estimated in the real or complex domain. cCSP's gains thus reflect the utility of operating on the full analytic signal (i.e., using amplitude and phase jointly, and the cross-electrode complex covariance structure they induce) instead of the magnitude summary that rCSP receives.

The former aligns with existing literature reporting gains from complex-valued CSP pipelines on motor imagery data (Akuthota et al., 2023; Malan & Sharma, 2022; Mishuhina & Jiang, 2021), and shows that the analytic signal carries discriminative information that amplitude alone underutilises. Once again, this is consistent with the theoretical argument that real-valued approaches systematically discard phase dependencies and cross-electrode relationships (Akuthota et al., 2023; Malan & Sharma, 2022).

Expanding TIMBRE's domain

After extensive hyperparameter optimization, TIMBRE (with batch normalization and L2 regularization) outperformed cCSP in the majority of subjects and configurations. Accuracy gains were modest in terms of magnitude, and depended on the subject. For subject al, tuned TIMBRE with 16 hidden nodes achieved a mean cross-validated accuracy of 0.782, compared to 0.754 for tuned CSP (16 features). The overlap between standard deviation from folds, indicates

the lack of substantial improvement. For subject aa, both methods converged near the same low ceiling (~ 0.57), and the advantage of added model complexity was absent, as demonstrated by the shape of the curves in figure 2. For subject aw, tuned TIMBRE produced gains over cCSP though absolute performance remained moderate.

The pattern across subjects suggests that TIMBRE’s additional expressive capacity is useful in some subjects, but not all. As a result, it is fair to say that the marginal gain from learned complex projections over analytically derived ones is highly dataset and subject specific. This could be interpreted as the first few cCSP filters capturing the complex-domain structure, which already take advantage of phase and amplitude jointly. TIMBRE’s hidden layer learns a similar class of features but via gradient descent on a more sensitive loss landscape, and the additional degrees of freedom it introduces do not consistently extract information that cCSP misses.

TIMBRE’s out-of-the-box performance relative to its design potential should not be seen as a generic failure of complex-valued neural networks. After all, TIMBRE was originally developed and validated on multi-electrode LFP recording from rat hippocampus, where oscillations are structured in a way that the carrier-free decoding problem was specifically designed to exploit (Agarwal et al., 2026). The scalp EEG motor imagery dataset at hand exists in a different regime, one with a lower signal-to-noise ratio, and smaller training data sizes relative to the whitened input. These factors mean that the architectural assumptions and empirical choices that made it effective in the LFP context may not translate without modification.

A note on optimization

The Optuna search conducted in this thesis served an analytical purpose beyond maximizing accuracy. Exploring a search space over regularization, normalization strategy, and fine-tuned hyperparameters such as patience, and learning rate, across all subjects and folds, characterized

the sensitivity and best model performance model capacity of TIMBRE within the current preprocessing pipeline.

The following findings emerged from this venture:

1. The optimizer consistently chose batch normalization and L2 regularization together across all 15 subject-fold combinations. One possible reason is that such a combination provides a more stable optimization trajectory in this setting than unit-norm, regardless of subject. The unit-norm constraint (present in the original TIMBRE), may impose too rigid a constraint on weight magnitude for a low SNR setting.
2. Having said that, minimal performance gains in subject aa (+0.017) compared to al (+0.075), for example, suggest that the best performing models for aa is a property of the data and the model rather than the search procedure. Extensive sampling of the hyperparameter space did not reveal configurations that broke past the previously observed performance ceiling for that subject.
3. Hyperparameter transferability analysis showed that configurations optimized on one subject produced near-zero accuracy changes, and even harmed performance in some cases, when applied to other subjects. Thus confirming that the gains from optimization are specific to each subject's data distribution, and discrediting the notion that there exists a single good configuration for this task. The fact that some subjects' Optuna landscapes prefer high learning rates with strong batch normalisation (e.g., al), while others (e.g., aa) tolerate or prefer the unit-norm constraint without L2, is consistent with subject-level differences in trial-to-trial signal stability. Subject al has the cleanest data in this pool (highest baseline accuracy and lowest fold-to-fold variance, see Figure 2); the gradient signal in such a regime is informative enough that batch normalisation's whitening of

layer activations and L2's stabilising effect can both be exploited. Subject aa's lower SNR may make the loss landscape too rugged for the additional flexibility to help, which is also consistent with the near-flat performance curve as a function of components for that subject.

All in all, these findings suggest that performance limits reached in this thesis reflect constraints of the models and dataset at hand. Within the explored feature space, both cCSP and TIMBRE have exploited what is recoverable from the analytic signal under 5-fold cross-validation.

Spectral Analysis

A substantial contribution of this work is the spectral analysis of learned model activations. Examining the power spectra of complex projections on held-out test data revealed an asymmetry in class-related activation across subjects. In all subjects and unit pairs, one class (Class 1, foot) was more strongly expressed in the learned projections than the other, leading to larger activation power across the relevant frequency range rather than equal sensitivity to both conditions. A two-way ANOVA revealed significant class-dependent differences in TIMBRE activations across specific frequency bands after the FDR (False Discovery Rate) correction. In conjunction, these findings indicate that the models do not respond symmetrically to both classes, and one class is more consistently encoded in the model's projections. Although statistical testing supports the reliability of this effect, its interpretation remains speculative, for lack of a better word. This effect might reflect a neurophysiological difference between the two conditions as recorded in these subjects, or the latent influence of artifacts in specific subjects. This analysis does unveil that accuracy alone would not reveal such a structure, as two classifiers with similar accuracy can differ substantially in the spectral content they rely on. A model whose activations concentrate power in the mu and beta bands is exploiting spectral structure

that aligns with the sensorimotor rhythms known to modulate during motor imagery (Pfurtscheller & Neuper, 1994), while a model that relies on broadband amplitude differences may be responding to structure that is harder to link to the relevant oscillations that motivated the use of complex-valued representations. The fact that both cCSP and TIMBRE show similar asymmetries suggests that class structure in this dataset is itself asymmetric, and that both methods pick up on the same signal despite doing so through different computational methods. One physiologically motivated reading of the foot over hand activation asymmetry is that the cortical generators of foot motor imagery (medial-frontal sensorimotor cortex along the longitudinal fissure) and hand motor imagery (lateral primary motor / pre-motor cortex) project to scalp electrodes with different spatial signatures. Because foot-imagery generators sit deeper and closer to midline, their projection to the scalp is more focal and dorsally distributed; the spatial filters learned by the projection layer may align more cleanly with this focal pattern than with the broader, more bilateral hand-imagery pattern, producing a higher-amplitude response on foot trials. An alternative reading is that the asymmetry reflects an idiosyncratic property of this dataset's labelling or instruction set rather than a general feature of motor imagery; replication on an independent dataset would be required to disambiguate.

Limitations

The demonstration that cCSP outperforms rCSP in this dataset is evidence that phase information in the analytic signal carries class-discriminative content. This decoding framework operates on correlates of neural activity at the scalp, filtered and projected through spatial filters; hence, the link from learned filter activations to specific neural populations is indirect and not established here.

The interpretability claims presented in this thesis are model-conditioned: the spectral profiles describe what the models have learned to respond to within the input analytic signal, given the preprocessing pipeline (high-pass filter at 2 Hz, Hilbert, train-fold whitening). Up to those preprocessing choices, the spectral analysis is model-free in the sense that the FFT is computed directly on the activations without further parametric assumptions; it therefore provides a description of the projection's frequency content that is reproducible from the trained model and the test data alone. What it does not do is characterise the underlying EEG or its neural generators directly, which lies outside the scope of this work. The findings remain relevant to this dataset because the same preprocessing applies across all subjects, and the cross-subject consistency of the foot > hand asymmetry is therefore a feature of the dataset under this pipeline. Finally, the analyses are specific to the three subjects from a single publicly available dataset (*Data Set IVa for the BCI Competition III*, 2003), with subject selection based on data availability rather than random selection. The generalizability of the quantitative findings cannot be assessed from this work alone, and further testing on different EEG datasets is required.

Conclusion

This work demonstrates that complex-valued representations improve EEG motor imagery decoding relative to real-valued methodologies. This improvement is interpretable in terms of the spectral content the models learn to exploit, as TIMBRE activations exhibit statistically significant asymmetric class encoding across frequency bands.

Back to CSP, cCSP outperforms rCSP using the same computational framework, and produces projections that can be examined in the frequency domain. TIMBRE also outperforms rCSP, but does not uniformly outperform cCSP even after extensive optimization. Spectral analysis of model activations revealed an asymmetry in model responsiveness between classes at a

statistically significant level where tested (TIMBRE only). Pragmatically, the most useful finding for EEG-based BCI applications may be that extending the CSP framework to the complex domain is both effective and interpretable without adding architectural complexity.

Code

All code for preprocessing, analysis, and figures can be found at

<https://github.com/gassyguillermomendoza/complex-valued-models-on-BCI-thesis>.

Acknowledgements

I am grateful to the members of BeatLab for their thoughtful feedback, and in particular to Xiaoyu Sun for the collaborative work on the cCSP implementation. I thank Professor Kohn for his careful reading of earlier drafts, and Professor Agarwal for his guidance and patience throughout.

And to my friends, grrrah¹ for two years of mocking me about my daughter (this thesis) whom I am at last sending out into the world.

¹ SpongeBob, big guy pants, okay.

Additional Figures

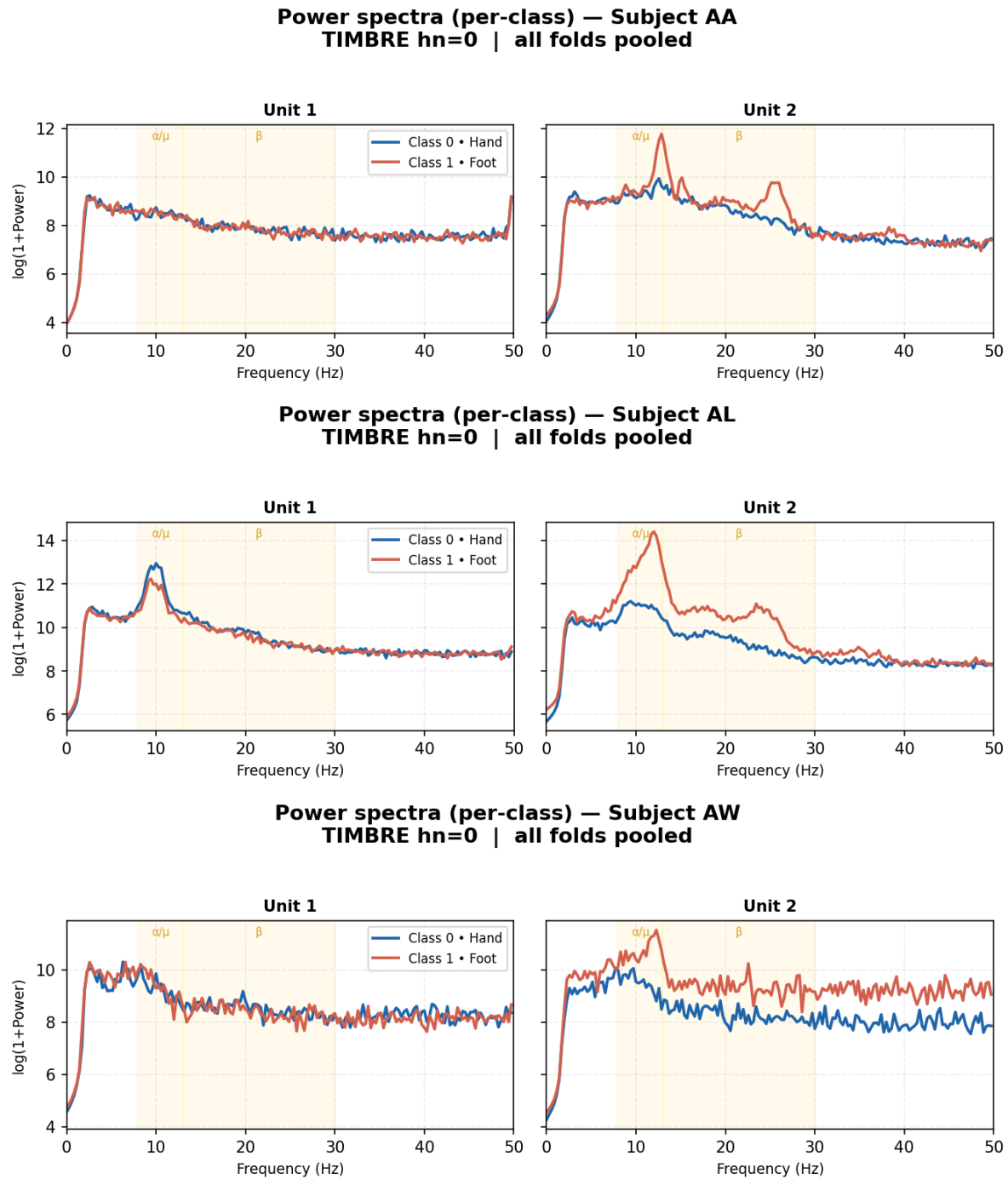


Figure A. Per-subject full power spectra for fixed-config TIMBRE (0 Hidden nodes) with mu and beta bands annotated.

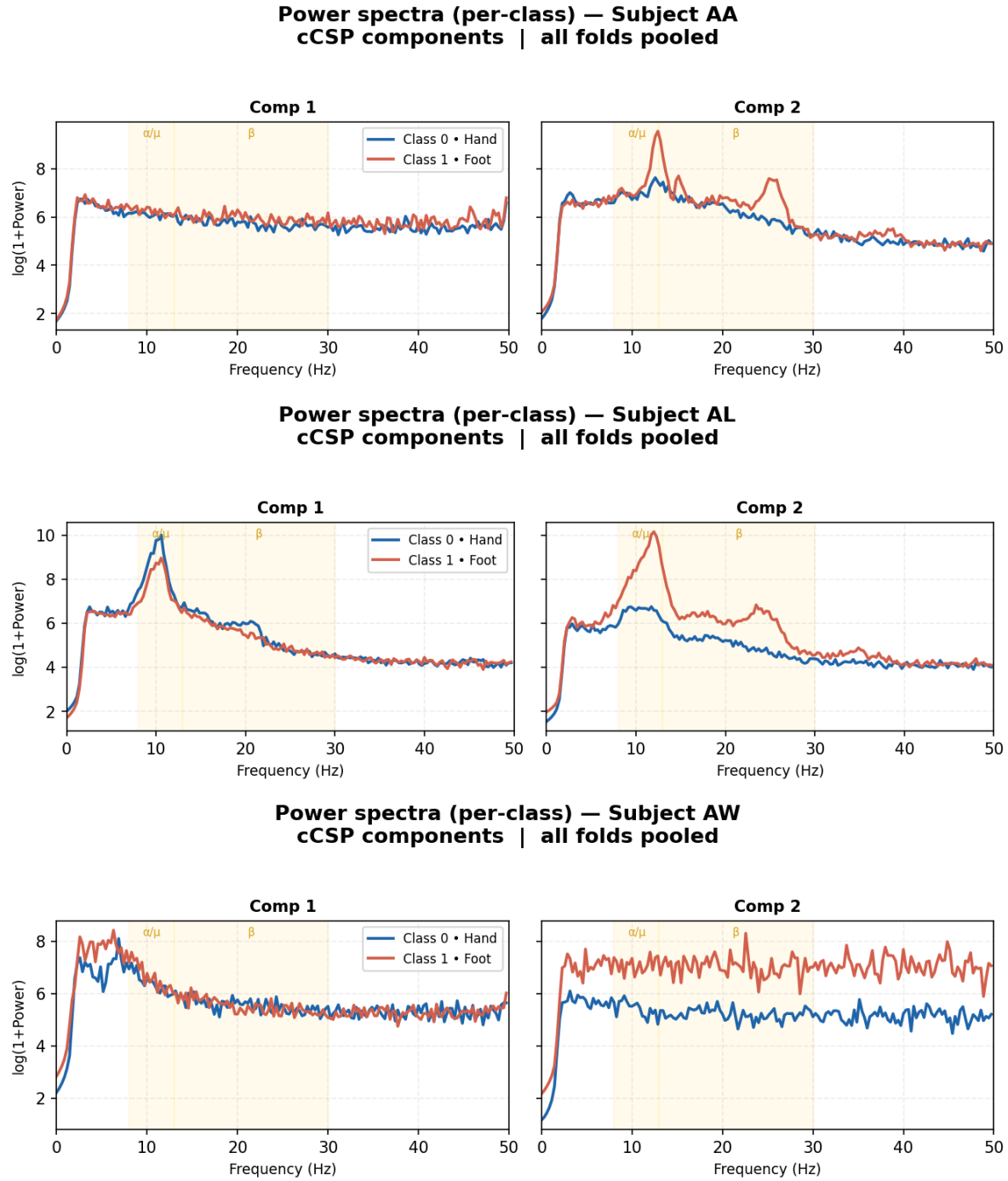


Figure B. Per-subject full power spectra for fixed-config cCSP (1 feature) with mu and beta bands annotated.

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